

2024 AMC 10A Problem 24 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 24

Problem:

A bee is moving in three-dimensional space. A fair six-sided die with faces labeled A^+ , A^- , B^+ , B^- , C^+ , and C^- is rolled. Suppose the bee occupies the point (a, b, c) . If the die shows A^+ , then the bee moves to the point $(a + 1, b, c)$, and if the die shows A^- , then the bee moves to the point $(a - 1, b, c)$. Analogous moves are made with the other four outcomes. Suppose the bee starts at the point $(0, 0, 0)$ and the die is rolled four times. What is the probability that the bee traverses four distinct edges of some unit cube?

Answer Choices:

- A. $\frac{1}{54}$
- B. $\frac{7}{54}$
- C. $\frac{1}{6}$
- D. $\frac{5}{18}$
- E. $\frac{2}{5}$

Solution:

Without loss of generality, assume that the first roll is A^+ . In order for the bee to traverse four distinct edges of a cube, the second roll cannot be A^+ or A^- , so there are 4 rolls (B^+ , B^- , C^+ , and C^-) that are allowed at this stage. Each new roll must represent a perpendicular direction for the bee, and there are 3 choices that remain in compliance for the third roll-2 of which extend into three dimensions, and 1 of which creates a "C" shape. In the former case, there are 2 choices for the fourth roll, while in the latter case, there are 3 choices (including the one where the bee traverses four edges forming a

square). In total, the number of compliant paths is $6 \cdot 4 \cdot (2 \cdot 2 + 1 \cdot 3) = 2^3 \cdot 3 \cdot 7$. The total number of paths is $6^4 = 2^4 \cdot 3^4$, and the probability that the path represents exactly four edges of a unit cube is

$$\frac{2^3 \cdot 3 \cdot 7}{2^4 \cdot 3^4} = \frac{7}{2 \cdot 3^3} = \text{(B)} \boxed{\frac{7}{54}}.$$

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